

What is the real key to physical AI? An event-driven camera? — and the lens behind it? (April 25, 2026)

Sony AI's ACE robot just beat professional table tennis players.

■ <https://lnkd.in/g-mzN3ex>

✦ The headline is reinforcement learning.

✦ The quiet enabler is something else:

◆ a hybrid of frame-based and event-based vision sensors delivering millimeter-accurate ball tracking at 200 Hz, spin at 700 Hz, and a 20.2 ms end-to-end loop — roughly $11\times$ faster than a top human's reaction.

✦ Without event cameras in that stack, the result is not possible.

So the question for the optics community:

✦ what is the lens design behind a physical-AI system?

■ perceives the world by change, not by frames?

✦ The answer reorders our priorities.

■ Mid-band MTF on moving edges replaces peak resolution.

■ Stray-light suppression ($PST < 1e-5$) becomes first-order

— a single ghost crossing the field is a continuous false-event storm.

■ Chief-ray angle tightens to match deep-well event pixels.

✦ The lens is no longer optimized for human image quality.

✦ It is optimized for task quality — the SNR of the events that drive the policy.

✦ The merit function shifts from Strehl to predicted event SNR over a moving-edge ensemble.

For working lens designers, this is not a theoretical reframing. It is a practical reset — tighter molds, sub-0.1% coatings, internal blackening, photon-counting QC at assembly. Even the BOM economics shift: NIR event optics can cost more than visible, with fewer elements. And more ...

■ And it goes further.

✦ The architectural pattern — perhaps as conventional CMOS plus an FPGA or ASIC front-end — extends naturally to NIR, SWIR, and LWIR: driver monitoring, AR/VR eye tracking, gas-leak detection, fire perimeter sensing, medical thermography.

✦ Each demands a new lens, designed against new merit functions.

■ But the deepest frontier is one wavelength — and one dimension — further.

✦ From classical to quantum: single-photon detectors are already event-driven by their physics — each photon is an asynchronous, time-tagged event.

✦ Couple them to neuromorphic processing, and the paradigm enters the quantum regime.

✦ Couple them to a dispersive element, and you have the event-driven spectrometer — a sparse stream of (wavelength, time) tokens

that turns chemistry, biology, and materials dynamics into something a robot can perceive in real time. Raman in living cells. Gas leaks at the photon limit. Real-time hyperspectral for agriculture, surgery, and remote sensing.

💠 Physical AI is not just a robotics story. It is an optics roadmap that runs from the visible to the single photon — across every wavelength, into the spectral domain, and from classical to quantum.

◆ Perhaps unsurprisingly: quantum photonics has been event-driven sensing and imaging from the start. We are only now learning to design the lenses for it.

#PhysicalAI #EventCamera #QuantumImaging #Spectroscopy #OpticalDesign #Photonics